

QTSR/RW – 24 / 1664

COMPUTER ARCHITECTURE

[Max. Marks : 60]

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Due credit will be given to neatness and adequate dimensions.
- (4) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) Explain steps involve processing an instruction in CPU and performance measures. 5(CO1)
 - (b) Implement the given function $F = \sum m(0, 1, 3, 5, 7)$ using PAL and PLA. 5(CO4)
2.
 - (a) Represent 85.125 (Decimal Number) into single precision IEEE 754 format. 5(CO2)
 - (b) Perform multiplication operation of -15 and -7 using Booth's algorithm. 5(CO2)
3. Design of Wilkes Control Unit for Multiplier. 10(CO5)
4.
 - (a) Assess the advantages and disadvantages of using cache memory in a computer. 5(CO1,3)
 - (b) Explain the concept of virtual memory and its role in memory management. 5(CO1,3)

5.
 - (a) Analysis of Direct Memory Access (DMA) in terms of their impact on CPU performance. 5(CO3)
 - (b) Discuss the different types of interrupts commonly used in modern systems. 5(CO1,3)

6.
 - (a) Discuss the significance of GPUs in modern computer architecture and identify recent trends in GPU development. 5(CO3)
 - (b) Compare and contrast the concepts of pipelining and parallel processing in terms of their impact on computer performance. 5(CO1)

